

## Solutions to Problem Set 8

### 1. Joint distribution and independence

- (a) Let  $A \subseteq \mathbb{R}^2$  be the closed triangle spanned by the points  $(0, 0)$ ,  $(0, 1)$  and  $(1, 0)$ . We say that the random vector  $(X, Y) : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$  follows a continuous uniform distribution on  $A$ , if the joint law of  $X, Y$ , denoted by  $\mathbf{P}_{(X,Y)}$ , has a multivariate probability density function

$$f_{X,Y}(x, y) = c \cdot \mathbb{1}_A(x, y) = \begin{cases} c, & (x, y) \in A, \\ 0, & (x, y) \notin A. \end{cases}$$

We write  $(X, Y) \sim \mathcal{U}(A)$ .

- i. Calculate the value of  $c$ . Explain briefly why the law of  $(X, Y)$  is called “uniform distribution”.
  - ii. Compute the marginal probability densities  $f_X$  and  $f_Y$ .
  - iii. Are  $X$  and  $Y$  independent? Give a precise mathematical argument.
  - iv. Calculate the covariance  $\text{Cov}[X, Y]$ .
- (b) Let  $X, Y \sim \mathcal{N}(0, 1)$  be independent random variables. We define the random variable  $Z$  by setting
- $$Z = \text{sign}(Y) \cdot X = \begin{cases} X, & \text{if } Y > 0 \\ -X, & \text{if } Y \leq 0. \end{cases}$$
- i. Determine the law of  $Z$ .
  - ii. Calculate the probability  $\mathbf{P}[X + Z = 0]$ . Is the random variable  $W = X + Z$  discrete, continuous or neither?
  - iii. Are  $X$  and  $Z$  independent? Give a precise mathematical explanation.

- (a) i. The triangle  $A$  spanned by the points  $(0, 0)$ ,  $(0, 1)$  and  $(1, 0)$  can be written as

$$A = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}.$$

We must have

$$\begin{aligned} 1 &\stackrel{!}{=} \iint_{\mathbb{R}^2} f_{X,Y}(x, y) d^2x = \int_0^1 \left( \int_0^{1-x} c \cdot dy \right) dx \\ &= c \int_0^1 (1-x) dx = c \left[ x - \frac{1}{2}x^2 \right]_0^1 = \frac{c}{2}. \end{aligned}$$

Therefore, we must have  $c = 2$ . Note that of course  $c = \frac{1}{|A|}$ , where  $|A|$  stands for the area measure of the set  $A$ . Now note that for any set  $B \subseteq \mathbb{R}^2$  (with  $B \in \mathcal{B}(\mathbb{R}^2)$ ), one has

$$\mathbf{P}_{(X,Y)}[B] = \mathbf{P}[(X, Y) \in B] = \int_B \frac{1}{|A|} \mathbb{1}_A(x, y) d^2x = \frac{|A \cap B|}{|A|}.$$

In other words, the probability of the point  $(X, Y)$  to fall into  $B$  is the ratio of the part of the measure of  $B$  in  $A$  and the measure of  $A$ . This justifies the name “uniform distribution” for  $\mathbf{P}_{(X,Y)}$ .

- ii. We see that

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_0^{1-x} 2 dy = 2(1-x),$$

whenever  $x \in [0, 1]$ , and 0 otherwise. Thus:

$$f_X(x) = 2(1-x) \mathbb{1}_{[0,1]}(x),$$

and by symmetry, we have

$$f_Y(y) = 2(1-y) \mathbb{1}_{[0,1]}(y).$$

iii. The random variables  $X$  and  $Y$  are not independent. Indeed,

$$\mathbf{P}\left[X \geq \frac{3}{4}, Y \geq \frac{3}{4}\right] = 2 \int_{\mathbb{R}^2} \underbrace{\mathbb{1}_{\{(x,y); x \in [0,1], y \in [0,1-x]\}} \cdot \mathbb{1}_{\{(x,y); x \geq \frac{3}{4}, y \geq \frac{3}{4}\}}}_{=0} d^2x = 0.$$

On the other hand, we have

$$\mathbf{P}\left[X \geq \frac{3}{4}\right] = 2 \int_{\frac{3}{4}}^1 (1-x) dx = 2 \left[ x - \frac{1}{2}x^2 \right]_{\frac{3}{4}}^1 = \frac{1}{16} = \mathbf{P}\left[Y \geq \frac{3}{4}\right].$$

So we have

$$\mathbf{P}[X \in [\frac{3}{4}, \infty), Y \in [\frac{3}{4}, \infty)] \neq \mathbf{P}[X \in [\frac{3}{4}, \infty)] \cdot \mathbf{P}[Y \in [\frac{3}{4}, \infty)].$$

iv. Let us calculate

$$\mathbf{E}[X] = \int_0^1 2x(1-x) dx = \left[ x^2 - \frac{2x^3}{3} \right]_0^1 = \frac{1}{3}.$$

By symmetry:

$$\mathbf{E}[Y] = \frac{1}{3}.$$

Now we have

$$\begin{aligned} \mathbf{E}[XY] &= \int_0^1 \left( \int_0^{1-x} 2xy dy \right) dx = \int_0^1 [xy^2]_0^{1-x} dx \\ &= \int_0^1 x(1-x)^2 dx = \int_0^1 (x + x^3 - 2x^2) dx \\ &= \left[ \frac{1}{2}x^2 + \frac{1}{4}x^4 - \frac{2}{3}x^3 \right]_0^1 = \frac{1}{2} + \frac{1}{4} - \frac{2}{3} = \frac{1}{12}. \end{aligned}$$

Therefore

$$\text{Cov}[X, Y] = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y] = \frac{1}{12} - \frac{1}{3} \cdot \frac{1}{3} = -\frac{1}{36}.$$

- (b) i. We calculate the cumulative distribution function of  $Z$ . Using the law of total probability, we obtain

$$\begin{aligned} F_Z(z) &= \mathbf{P}[Z \leq z] = \mathbf{P}[Z \leq z | Y > 0] \mathbf{P}[Y > 0] + \mathbf{P}[Z \leq z | Y \leq 0] \mathbf{P}[Y \leq 0] \\ &= \mathbf{P}[X \leq z | Y > 0] \cdot \frac{1}{2} + \mathbf{P}[-X \leq z | Y \leq 0] \cdot \frac{1}{2} \\ &\stackrel{\text{indep.}}{=} \frac{1}{2} \mathbf{P}[X \leq z] + \frac{1}{2} \mathbf{P}[-X \leq z] \\ &= \frac{1}{2} \Phi(z) + \frac{1}{2} \Phi(z) = \Phi(z), \end{aligned}$$

where we defined  $\Phi(u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^u e^{-\frac{1}{2}x^2} dx$  (the cumulative distribution function of a  $\mathcal{N}(0, 1)$ -distribution), and used the fact that for  $X \sim \mathcal{N}(0, 1)$ , also  $-X \sim \mathcal{N}(0, 1)$  (see Example 5.6, (ii)).

- ii. Note that  $\mathbf{P}[2X = 0] = 0$ . Therefore, by definition, we see that

$$\mathbf{P}[X + Z = 0] = \mathbf{P}[Z = -X] = \mathbf{P}[Y \leq 0] = \frac{1}{2}.$$

In particular,  $W = X + Z$  is not continuous (since for a continuous random variable  $U$  one has  $\mathbf{P}[U = c] = 0$  for every  $c \in \mathbb{R}$ ). Moreover,  $W$  is not discrete, since it can attain overcountably many values (namely all of the values in  $\mathbb{R}$ ).

- iii.  $X$  and  $Z$  are not independent. Indeed, if  $X$  and  $Z$  were independent, then we would have for every  $A, B \in \mathcal{B}(\mathbb{R})$  that

$$\mathbf{P}[X \in A, Z \in B] = \mathbf{P}[X \in A] \cdot \mathbf{P}[Z \in B].$$

This is not fulfilled for instance for  $A = [-1, 1]$  and  $B = [2, \infty)$ , since

$$\mathbf{P}[X \in [-1, 1], Z \in [2, \infty)] = 0 \neq \underbrace{\mathbf{P}[X \in [-1, 1]]\mathbf{P}[Z \geq 2]}_{>0}.$$

## 2. (\*) Bertrand's paradox

[4 Points]

We consider an equilateral triangle inscribed in a circle of radius 1. Suppose a chord of the circle is chosen at random. What is the probability that the chord is longer than a side of the triangle? For solving this try the following probability models:

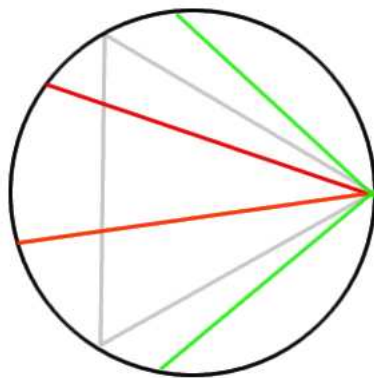
- (a) The “random endpoints” method: Choose two uniform random points on the circumference of the circle and draw the chord joining them, i.e., let  $U, V \sim \mathcal{U}((0, 1))$ , define

$$X = e^{U2\pi i} = (\cos(2\pi U), \sin(2\pi U)), \quad Y = e^{V2\pi i} = (\cos(2\pi V), \sin(2\pi V))$$

and take the chord connecting  $X$  and  $Y$ .

- (b) The “random radius” method: Choose a radius of the circle, choose a uniform point on the radius and construct the chord through this point and perpendicular to the radius, i.e., choose a radius and choose  $R \sim \mathcal{U}((0, 1))$ , take the point on the radius that is at distance  $R$  from the center and the chord will be the only chord perpendicular to this point.
- (c) The “random midpoint” method: Choose a point uniformly anywhere within the circle (see Problem 1.(a), but with  $A$  replaced by a disk) and construct a chord with the chosen point as its midpoint, i.e., and take the chord whose midpoint is  $(X, Y)$ .
- (d) Is this a contradiction?

- (a) Remember that  $x \bmod 1 = x - \lfloor x \rfloor$ . Note that the chord is longer than a side of the triangle if  $(V - U) \bmod 1 \in (\frac{1}{3}, \frac{2}{3})$ . This is because the chord going from 1 to  $f(V - U)$  has the same length that going from  $X$  to  $Y$  (we just did a rotation). For the chord to be longer than the side of a triangle we need: That  $F(V - U)$  is a red line, that means that  $(V - U) \bmod 1 \in (\frac{1}{3}, \frac{2}{3})$ .



Additionally :

$$\begin{aligned}
\mathbf{E} \left[ \mathbf{1}_{\{(V-U) \bmod 1 \in (\frac{1}{3}, \frac{2}{3})\}} \right] &= \iint_{(0,1)^2} \mathbf{1}_{\{(y-x) \bmod 1 \in (\frac{1}{3}, \frac{2}{3})\}} dy dx \\
&= \int_0^1 \left( \int_0^1 \mathbf{1}_{\{y \in (\frac{1}{3}, \frac{2}{3})\}} dy \right) dx \\
&= \int_0^1 \frac{1}{3} dy \\
&= \frac{1}{3}.
\end{aligned}$$

- (b) Let  $r$  be the point chosen in the radius, it is a uniformly distributed random variable over  $[0, 1]$ . The length of the chord will be given by  $2\sqrt{(1-r^2)}$ . Then

$$\begin{aligned}
\mathbf{P} \left[ 2\sqrt{(1-r^2)} \geq \sqrt{3} \right] &= \mathbf{P} \left[ 1-r^2 \geq \frac{3}{4} \right] \\
&= \mathbf{P} \left[ r^2 \leq \frac{1}{4} \right] \\
&= \frac{1}{2}.
\end{aligned}$$

- (c) Let  $(x, y)$  be the point chosen in the circle, it is a uniform random variable over the disk  $B(0, 1) = \{(x, y) : x^2 + y^2 < 1\}$ . The length of the chord will be given by  $2\sqrt{1-(x+y)^2}$ . Thus,

$$\begin{aligned}
\mathbf{P} \left[ 2\sqrt{(1-(x+y)^2)} \geq \sqrt{3} \right] &= \mathbf{P} \left[ 1-(x+y)^2 \geq \frac{3}{4} \right] \\
&= \mathbf{P} \left[ (x+y)^2 \leq \frac{1}{4} \right] \\
&= \iint_{B(0, \frac{1}{2})} \frac{1}{\pi} dx dy \\
&= \frac{1}{4}.
\end{aligned}$$

- (d) This is not a contradiction. What this show us that there is not a single canonical way to “pick a chord uniformly at random” in this circle, so we have to define the probability measure we are interested in before asking the question about the probability of an event.

### 3. Convolution

- (a) On normal distributions:

- i. Let  $(\Omega, \mathcal{F}, \mathbf{P})$  be a probability space and  $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$  independent random variables with  $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$  and  $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$  with  $\mu_1, \mu_2 \in \mathbb{R}$  and  $\sigma_1, \sigma_2 > 0$ . Show that

$$X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

*Hint:* First explain why it suffices to show this for  $\mu_1 = \mu_2 = 0$  and  $\sigma_1 = 1$ . Then complete the square in the exponent.

- ii. Now let  $n \in \mathbb{N}$  and let  $X_1, \dots, X_n : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$  i.i.d. random variables with  $X_1 \sim \mathcal{N}(\mu, \sigma^2)$ , where  $\mu \in \mathbb{R}$  and  $\sigma > 0$ . Consider the mean value  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . Determine the law of the random variable

$$Z_n = \sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}.$$

- (b) On Poisson distributions:

- i. Let  $(\Omega, \mathcal{F}, \mathbf{P})$  be a probability space and  $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$  independent random variables with  $X \sim \text{Pois}(\lambda)$  and  $Y \sim \text{Pois}(\mu)$ , where  $\lambda, \mu > 0$ . Show that

$$X + Y \sim \text{Pois}(\lambda + \mu).$$

- ii. Give a concrete counterexample showing that  $X + Y$  does in general *not* follow a Poisson distribution, if we drop the assumption of independence in part i. of the problem.

- (a) i. We first argue why it suffices to consider the case where  $\mu_1 = \mu_2 = 0$  and  $\sigma_1 = 1$ . Recall that for  $W \sim \mathcal{N}(\mu, \sigma^2)$ , and  $a, b \in \mathbb{R}$ ,  $a \neq 0$ , one has  $aW + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$  (see (5.18) in the notes). Now we can write

$$X + Y = \sigma_1 \left( \frac{X - \mu_1}{\sigma_1} + \frac{Y - \mu_2}{\sigma_1} \right) + \mu_1 + \mu_2.$$

The random variables  $X' = \frac{X - \mu_1}{\sigma_1} \sim \mathcal{N}(0, 1)$  and  $Y' \sim \mathcal{N}(0, \frac{\sigma_2^2}{\sigma_1^2})$  are independent (Proposition 7.15). If we can show that

$$(\star) \quad X' + Y' \sim \mathcal{N}\left(0, 1 + \frac{\sigma_2^2}{\sigma_1^2}\right),$$

we find that

$$X + Y = \sigma_1(X' + Y') + \mu_1 + \mu_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Let us therefore show  $(\star)$  and set  $\sigma = \frac{\sigma_2}{\sigma_1}$ . We use the formula for the sum of independent continuous random variables (Proposition 7.25). Now

$$f_{X'}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, \quad f_{Y'}(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}y^2}.$$

For  $Z = X' + Y'$  we have

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X'}(z - y) f_{Y'}(y) dy = \frac{1}{2\pi\sigma} \int_{-\infty}^{\infty} e^{-\frac{(z-y)^2}{2}} e^{-\frac{y^2}{2\sigma^2}} dy.$$

We see that

$$\begin{aligned} \frac{y^2}{\sigma^2} + (z - y)^2 &= \frac{y^2}{\sigma^2} + z^2 - 2yz + y^2 \\ &= \left(1 + \frac{1}{\sigma^2}\right) \cdot \left(y^2 - 2y \frac{z}{1 + \frac{1}{\sigma^2}} + \left(\frac{z}{1 + \frac{1}{\sigma^2}}\right)^2\right) + z^2 \left(1 - \frac{1}{1 + \frac{1}{\sigma^2}}\right) \\ &= \left(1 + \frac{1}{\sigma^2}\right) \cdot \left(y - \frac{z}{1 + \frac{1}{\sigma^2}}\right)^2 + \frac{z^2}{1 + \sigma^2}. \end{aligned}$$

With this, we find

$$\begin{aligned} f_Z(z) &= \frac{(1 + \frac{1}{\sigma^2})^{-1/2}}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi(1 + \frac{1}{\sigma^2})^{-1}}} e^{-\frac{1}{2(1 + \frac{1}{\sigma^2})^{-1}} \left(y - \frac{z}{1 + \frac{1}{\sigma^2}}\right)^2} e^{-\frac{z^2}{2(1 + \sigma^2)}} dy \\ &= \frac{1}{\sqrt{2\pi(1 + \sigma^2)}} e^{-\frac{z^2}{2(1 + \sigma^2)}}. \end{aligned}$$

Here we used that

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi(1 + \frac{1}{\sigma^2})^{-1}}} e^{-\frac{1}{2(1 + \frac{1}{\sigma^2})^{-1}} \left(y - \frac{z}{1 + \frac{1}{\sigma^2}}\right)^2} dy = 1,$$

since this is the integral of the density of a  $\mathcal{N}(\frac{z}{1 + \frac{1}{\sigma^2}}, (1 + \frac{1}{\sigma^2})^{-1})$ -distribution. We thus find that  $Z \sim \mathcal{N}(0, 1 + \sigma^2)$ , which is  $(\star)$ .

ii. Note that by (a), we have

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n^2\sigma^2) \quad \Rightarrow \quad \overline{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

In the last step, we used again (5.18) from the notes. Now

$$\overline{X}_n - \mu \sim \mathcal{N}\left(0, \frac{\sigma^2}{n}\right) \quad \Rightarrow \quad \frac{\sqrt{n}}{\sigma}(\overline{X}_n - \mu) \sim \mathcal{N}(0, 1).$$

- (b) i. We use the formula for the probability mass function of independent discrete real random variables (Proposition 7.23). Remember that  $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}$ , for  $k \in \mathbb{N}_0$ ,  $p_Y(\ell) = \frac{\mu^\ell}{\ell!} e^{-\mu}$ . Therefore, for  $Z = X + Y$  (and by setting  $p_X(n) = 0$  for  $n < 0$ ):

$$\begin{aligned} p_Z(k) &= \sum_{\ell \in \mathbb{N}_0} p_X(k-\ell) p_Y(\ell) = \sum_{\ell=0}^k \frac{\lambda^{k-\ell}}{(k-\ell)!} e^{-\lambda} \frac{\mu^\ell}{\ell!} e^{-\mu} \\ &= \frac{e^{-(\lambda+\mu)}}{k!} \sum_{\ell=0}^k \underbrace{\frac{k!}{(k-\ell)! \ell!} \mu^\ell \lambda^{k-\ell}}_{=(\mu+\lambda)^k} = \frac{(\mu+\lambda)^k}{k!} e^{-(\lambda+\mu)}. \end{aligned}$$

- ii. Let  $Y = X \sim \text{Pois}(\lambda)$ . Then we have that  $Z = X + Y = 2X$  is not Poisson distributed. Indeed, assume that  $Z \sim \text{Pois}(\mu)$  for some parameter  $\mu > 0$ . Then

$$\mathbf{P}[Z = 1] = \mu e^{-\mu}.$$

However,

$$\mathbf{P}[Z = 1] = \mathbf{P}[2X = 1] = \mathbf{P}[X = \frac{1}{2}] = 0,$$

a contradiction.

#### 4. (\*) The student- $t$ -distribution

[4 Points]

Suppose that  $Z$  has the standard normal distribution,  $V$  has the  $\chi^2$  distribution with  $n$  degrees of freedom, and that  $Z$  and  $V$  are independent. The  $\chi^2$ -distribution with  $n$  degrees of freedom is defined as the distribution of  $\sum_{i=1}^n X_i^2$  where  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ . Let

$$T = \frac{Z}{\sqrt{V/n}}.$$

You will show that  $T$  has p.d.f. given by

$$f(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2} \quad t \in \mathbb{R},$$

where we define  $\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$ , for  $t > 0$ .

- (a) Show that  $V$  has the probability density function  $f_V(x) = c_n x^{n/2-1} e^{-x/2} \mathbb{1}_{\{x \geq 0\}}$ , with an appropriate  $c_n \in \mathbb{R}$ .  
*Hint:* Calculate  $c_n$  first, then use induction.
- (b) Find the joint probability density function of  $(T, V)$ .  
*Hint:* As always, look at the (joint) cumulative distribution function  $F_{T,V}(t, v)$ .
- (c) Find the probability density function of  $T$ .

- (a) We let  $V_n = \sum_{i=1}^n X_i^2$ . We first show the claim for  $n = 1$ . Indeed, this is the  $\chi^2$ -distribution with one degree of freedom, and we know from the notes (formula (5.24)) that

$$f_{V_1}(x) = \frac{1}{\sqrt{2\pi}} x^{-\frac{1}{2}} e^{-\frac{x}{2}} \mathbb{1}_{(0, \infty)}(x),$$

which is the claimed formula with  $c_1 = \frac{1}{\sqrt{2\pi}}$ .  
Now let  $n > 1$ , and take  $x > 0$ , then

$$\begin{aligned} f_{V_{n+1}}(x) &= \int_0^x f_{V_n}(y) f_{X_{n+1}^2}(x-y) dy \\ &= \int_0^x c_n y^{n/2-1} e^{-y/2} c (x-y)^{-1/2} e^{-(x-y)/2} dy \\ &= c c_n e^{-x/2} \int_0^x y^{n/2-1} (x-y)^{-1/2} dy \\ &= c c_n e^{-x/2} \int_0^1 (xz)^{n/2-1} (x(1-z))^{-1/2} x dz \\ &= c c_n e^{-x/2} x^{(n+1)/2-1} \int_0^1 z^{n/2-1} (1-z)^{-1/2} dz, \end{aligned}$$

where  $c = \frac{1}{\sqrt{2\pi}}$ . The last integral is independent of  $n$ , so we proved the claim with

$$c_{n+1} = c c_n \int_0^1 z^{n/2-1} (1-z)^{-1/2} dz.$$

To find the value of  $c_n$ , we simply note that  $f_V$  needs to be a density, so

$$\int_0^\infty c_n x^{n/2-1} e^{-x/2} dx = \int_0^\infty 2c_n (2z)^{n/2-1} e^{-z} dz = 2^{\frac{n}{2}} c_n \Gamma\left(\frac{n}{2}\right),$$

which means that  $c_n = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})}$ .

- (b) We use the independence of  $Z$  and  $V := X_1^2 + \dots + X_n^2$  and see that the joint probability density function of  $Z$  and  $V$  for  $x \in \mathbb{R}, z > 0$  is given by

$$f_{Z,V}(z, v) = \frac{c_n}{\sqrt{2\pi}} e^{-x^2/2} e^{-z/2} z^{n/2-1}.$$

We now find the joint probability density function of  $(T, V)$ . To that end, note that

$$\begin{aligned} F_{T,V}(t, v) &= \mathbf{P}[T \leq t, V \leq v] = \mathbf{P}[Z \leq t\sqrt{V/n}, V \leq v] \\ &= \int_0^\infty f_V(v) \left( \int_{-\infty}^{t\sqrt{v/n}} f_Z(z) dx \right) dv, \end{aligned}$$

having used independence of  $Z$  and  $V$ . Taking partial derivatives in  $v$  and  $z$ , we see that the joint probability density function is

$$f_{T,V}(t, v) = f_Z(t(v/n)^{1/2})(v/n)^{1/2} f_V(v)$$

for  $t \in \mathbb{R}, v > 0$ .

- (c) By integration over  $v$  we obtain

$$\begin{aligned} f_T(t) &= \int_0^\infty f_{T,V}(t, v) dv = \int_0^\infty f_Z(t(v/n)^{1/2})(v/n)^{1/2} f_V(v) dv \\ &= \int_0^\infty \frac{c_n}{\sqrt{2\pi}} e^{-\frac{t^2 v}{2n}} e^{-v/2} v^{n/2-1} (v/n)^{1/2} dv \\ &= \frac{c_n}{\sqrt{2\pi n}} \int_0^\infty \exp(-v(1 + t^2/n)/2) v^{((n+1)/2)-1} dv. \end{aligned}$$

Using the substitution  $u = v(1 + t^2/n)/2$  this becomes

$$f_T(t) = \frac{c_n}{\sqrt{2\pi n}} \left( \frac{2}{1 + t^2/n} \right)^{(n+1)/2} \underbrace{\int_0^\infty e^{-u} u^{(n+1)/2-1} du}_{\Gamma(\frac{n+1}{2})}.$$

Therefore, we see that

$$f_T(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n} \Gamma(\frac{n}{2})} \left( \frac{1}{1 + t^2/n} \right)^{(n+1)/2}, \quad t \in \mathbb{R}.$$